

MULTI-VENDOR BOOK STORE

Project Use Case & Solution Document

React.js • JavaScript • Redux Toolkit • Micro Frontend

Document Type	Project Use Case Document
Application	Multi-Vendor Book Store
Frontend	React.js with JavaScript
Architecture	Micro Frontend (MFE) + Redux Toolkit

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1. Project Overview

The Multi-Vendor Book Store is a digital marketplace that allows multiple independent vendors to sell books through a single platform. Customers can discover books, compare vendors, add items to a cart, place orders and track deliveries. Vendors can manage their catalog, inventory and orders, while administrators control the overall marketplace.

2. Business Objectives

- Provide a single marketplace for multiple book vendors.
- Allow customers to easily discover and purchase books.
- Give vendors an independent portal to manage books, stock and sales.
- Provide administrators with centralized marketplace control.
- Use Micro Frontend architecture so business modules can be developed and deployed independently.
- Use Redux Toolkit for predictable and reusable application state management.

3. Technology Stack

Technology	Purpose
Frontend	React.js with JavaScript
State Management	Redux Toolkit
Routing	React Router
Micro Frontend	Webpack Module Federation
Backend	Node.js / Express
API	REST API
Database	MongoDB
UI	Material UI / CSS

4. User Roles

Role	Responsibilities
Customer	Browse, search, purchase books, manage cart/wishlist and track orders.
Vendor	Manage books, inventory, orders and sales.
Admin	Manage vendors, customers, books, orders, categories and reports.

5. Micro Frontend Architecture

The application is divided into independent business-focused Micro Frontends:

Host / Shell MFE – Authentication, common layout, navigation and application-level routing.

Customer MFE – Book catalog, search, details, cart and wishlist.

Vendor MFE – Vendor dashboard, book management, inventory and sales.

Admin MFE – Vendor/customer management, book management and reports.

Order MFE – Checkout, payment, order history and tracking.

6. Customer Use Cases

ID	Use Case	Description
UC-01	Customer Registration	Customer enters required details; the system validates the information and creates an account.
UC-02	Customer Login	Customer enters credentials; after successful authentication, the customer is routed to the dashboard.
UC-03	Browse Books	Customer views available books with title, author, price, vendor, rating and stock.
UC-04	Search Books	Customer searches books by title, author, ISBN, vendor or category.
UC-05	Filter & Sort	Customer filters by category, price, rating and availability and sorts results.
UC-06	View Book Details	Customer views description, ISBN, price, discount, vendor, reviews, rating and stock.
UC-07	Add to Cart	Customer adds a book to the cart; Redux cart state is updated.
UC-08	Wishlist	Customer adds or removes books from the wishlist and can move an item to the cart.
UC-09	Checkout	Customer enters delivery and payment information; the system calculates the final amount.
UC-10	Place Order	After successful payment, the system creates the order and notifies the relevant vendor.
UC-11	Track Order	Customer views order progress from confirmation through delivery.

7. Vendor Use Cases

ID	Use Case	Description
UC-12	Vendor Registration	Vendor submits business and contact details for administrator approval.
UC-13	Vendor Login	Approved vendor authenticates and accesses the vendor

		dashboard.
UC-14	Add Book	Vendor adds title, author, ISBN, category, description, price, discount, stock and image.
UC-15	Manage Books	Vendor can add, edit, delete, enable/disable books and update price or stock.
UC-16	Inventory Management	Vendor monitors available, low-stock and out-of-stock books.
UC-17	Order Management	Vendor accepts orders and updates them through packing, shipping and delivery.
UC-18	Sales Dashboard	Vendor views orders, books sold, sales and revenue.

8. Admin Use Cases

ID	Use Case	Description
UC-19	Admin Login	Admin authenticates and accesses administration modules.
UC-20	Vendor Management	Admin approves, rejects, suspends or activates vendors.
UC-21	Customer Management	Admin views, activates or blocks customer accounts.
UC-22	Book Management	Admin reviews books, manages categories and removes inappropriate listings.
UC-23	Order Management	Admin views platform-wide orders, payments, vendors and statuses.
UC-24	Reports	Admin views customer, vendor, book, order, revenue and category-wise reports.

9. End-to-End Business Flow

1. Customer Login
2. Search and select a book
3. View book details and vendor
4. Add book to cart
5. Checkout
6. Payment
7. Order creation
8. Vendor receives order
9. Vendor accepts and packs order

10. Vendor ships order
11. Customer tracks shipment
12. Order delivered

10. Redux Toolkit State Management

Redux Slice	Managed State
authSlice	user, role, token, authentication state
bookSlice	books, search text, filters and selected book
cartSlice	cart items, quantities and total amount
wishlistSlice	wishlist items
vendorSlice	vendor details and vendor dashboard state
orderSlice	orders, selected order and order status

11. Functional Requirements

- Customer: registration, login, search, filters, book details, cart, wishlist, checkout, payment and order tracking.
- Vendor: registration, admin approval, book management, inventory management, order management and sales dashboard.
- Admin: vendor management, customer management, book/category management, order management and reports.

12. Non-Functional Requirements

- Responsive and user-friendly interface.
- Secure authentication and role-based authorization.
- Independent MFE development and deployment.
- Reusable React.js components.
- Centralized Redux Toolkit state management.
- Reliable API error handling and validation.
- Scalable and maintainable architecture.
- Logging and monitoring support.

13. JavaScript MFE Project Structure

- bookstore-host/ – Host/Shell React JavaScript application
- customer-mfe/ – Customer React JavaScript application
- vendor-mfe/ – Vendor React JavaScript application
- admin-mfe/ – Admin React JavaScript application
- order-mfe/ – Order React JavaScript application
- shared/ – Redux Toolkit store and reusable components

14. Future Enhancements

- Online payment gateway integration.
- Book reviews and ratings.
- Vendor commission calculation.
- Email and SMS notifications.
- Personalized book recommendations.
- Advanced analytics dashboard.
- Coupon and discount management.